Before: computational problem → efficient algorithm

ex [Is there a cyclic reference in the cells of a spreadsheet?]

→ alg: reduced to detecting cycles in a dir. graph

ex is t reachable from s in G?

[→ alg: BFS $O(V+E)$]

alg: upper bound on hardness

now: computational problem → no efficient algorithm (probably)

lower bounds on hardness

Types of computational problems

① Decision problems: output yes/no

ex both probs above

Are there at least 2 paths of distinct lengths in G?

Is there a path of at least length k in G?

② Optimization problems: output the best numerical value

what is the length of a shortest path from s to t in G?

what is the length of the longest path in G?

③ Search problems: identify a particular object

a

what is the longest path in G?

w/ table: what is the optimization version of ✱?

decision version of ✱?

↳ can you make a few diff ones?

Example Problems

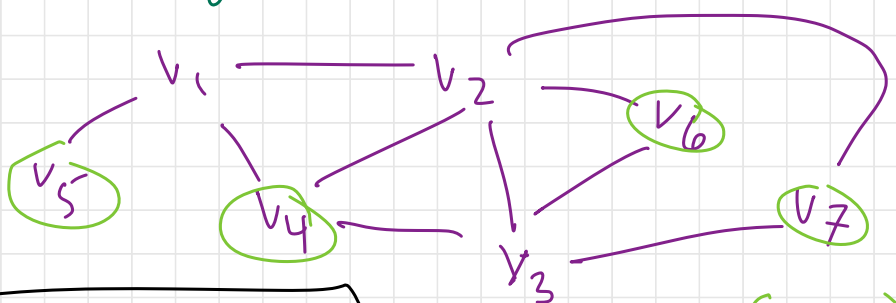
(1) Independent Set, IS

Given an undirected graph $G = (V, E)$

$X \subseteq V$ is an *independent set* if (↑ subset of)

no pair of nodes $u, v \in X$ is connected by an edge $(u, v) \in E$.

$G =$



is $X = \{v_5, v_4\}$ an IS? yes — is $(v_5, v_4) \in E$? no

$X = \{v_2, v_5, v_6\}$ an IS? $v_2 v_5$? no — no

$v_2 v_6$ yes

$X = \emptyset$ an IS? yes

search

What is the largest IS in G?

→ $X = \{v_5, v_6, v_7, v_4\}$

NP is the set of all decision problems that can be verified in polynomial time.

verified: if you propose a "yes" answer, you can check it.

ex problem: Is there a IS in G of size at least k?

verification algorithm (G, certificate)
input ↓ "yes" answer: X ↓

- does X have at least k elements? ↪ linear
- is X a valid IS?
  - for every pair of nodes u, v in X:
    - if (u,v) ∉ E:
      - return F
  - if all pass, return T

quad.

ex prob: Is k the size of the largest IS in G?

verification (G, certificate) ???